AASHINI RAJPAL

RESEARCH ENGINEER

Chemist/physicist with a strong foundation in experimental design, spectroscopy, microfluidics, dosimetry and process optimization

Portfolio

+33 (0) 767759893

Paris (Relocating to Berlin)

aashini.rajpal11@gmail.com

linkedin.com/in/aashinirajpal/

SKILLS

- Simulations (*GAUSSIAN, CPMD, BOMD, COMSOL, Chemsimul*)
- Spectroscopy techniques
- Microfluidics
- Proposal Writing
- Analysis softwares *(IGOR pro, CasaXPS, SpecsLab Prodigy, OxyGEN, ChemDraw, OceanView, Avantes*)
- Python (Intermediate)

EDUCATION

2020 - 2023
PhD in Chemistry
University Paris-Saclay

2018 - 2020
Master in Physical Chemistry
Erasmus Mundus SERP+ Joint Master Degree, France and Poland (16.5/20)

LANGUAGES

English

Hindi

French

REFERENCE

CH.ROBERT@gustaveroussy.fr
gerard.baldacchino@cea.fr
jean-philippe.renault@cea.fr
christophe.nicolas@synchrotron-soleil.fr

WORK EXPERIENCE

Gustave Roussy Cancer Institute and Theryq — 2024 - PRESENT
Research Engineer
- Performed dosimetric characterization under ultra-high dose-rate (UHDR) conditions using multiple techniques.
- Designed and optimized experimental workflows to investigate the FLASH/UHDR effect.
- Investigated chemistry-driven mechanisms underlying differential UHDR biological responses.
- Conducted heat diffusion simulations using COMSOL.
- Collaborated on projects integrating chemistry, physics, and radiobiology.

SOLEIL synchrotron, Sorbonne University and CEA Saclay — 2020 - 2023
Research scholar
- Developed and optimized a highly sensitive in-line analysis setup integrated with microfluidic devices.
- Investigated light-matter interaction using experiments and simulations. Worked with electron and photon radiation sources.
- Led a team of 5 researchers for 10 intensive experimental projects.
- Presented research findings at national and international conferences.

Indian Institute of Technology Bombay (IIT B) — 2017
Research scholar
- Computationally studied reaction mechanisms to analyze activation energies and molecular structures

CERTIFICATIONS

- **Good Clinical Practices**, National Institute on Drug Abuse (NIDA) Center for Clinical Trials (CCTN) Clinical Trials Network (CTN)
- **Project designer - Regulatory course on animal experiment**, approved by the Ministry of Agriculture
- **Molecule to Market job simulation**, Pfizer UK.
- **Basics of AI in Medical Physics ,** Gustave Roussy

PUBLICATIONS

Researchgate/aashinirajpal